

ADDITION & SUBTRACTION

Assessment Skill Taxonomy

A complete case-based framework for building diagnostic and Olympiad-style assessments

Purpose. This document treats a skill as a distinct assessment case whenever a child can make a different kind of mistake. Therefore horizontal vs vertical presentation, unequal digit lengths, number order, place-value alignment, carry/borrow location, zeros, answer-size changes, and missing digits are all tracked explicitly.

Scope: whole-number addition and subtraction. The framework is not tied to a specific grade and scales from 1-digit to N-digit numbers.

1. How to Read This Taxonomy

Dimension	Typical values	Why it matters
Operation	Addition / Subtraction	Different inverse relationships and regrouping logic.
Digits in first number	1 / 2 / 3 / 4 / N	Controls magnitude and place-value complexity.
Digits in second number	1 / 2 / 3 / 4 / N	Unequal lengths create alignment errors.
Order	Longer first / Shorter first	For addition, commutativity does not remove the diagnostic value of order.
Presentation	Vertical / Horizontal	Horizontal questions also test conversion and place-value alignment.
Regrouping	None / Single / Multiple	Separates basic calculation from carry/borrow proficiency.
Regroup location	Ones / Tens / Hundreds / ...	A child may succeed in one column but not another.
Pattern	Isolated / Consecutive / Non-adjacent / Cascading	Different carry/borrow chains expose different misconceptions.
Zeros	None / Internal zero / Across zero / Multiple zeros	Zeros substantially change borrowing/carrying demands.
Answer size	Same digits / Fewer digits / More digits	Tests whether the learner anticipates magnitude changes.
Unknown	None / Whole number / Digit / Multiple digits / Operation	Separates direct computation from reverse reasoning.
Question type	Direct / Word problem / Balance / Error diagnosis / Estimation	Measures calculation, application and reasoning separately.

Important: one question can carry several tags. Example: $345 + 7 = ?$ is Addition + 3-digit/1-digit + horizontal + unequal length + longer-first + no-carry (or carry, depending on digits).

2. Addition - Direct Calculation Skills

2.1 Single-digit + Single-digit

ID	Skill / Case	Example
A01	Vertical; answer remains 1 digit	$3 + 4 = 7$
A02	Vertical; answer becomes 2 digits	$7 + 8 = 15$
A03	Horizontal; answer remains 1 digit	$3 + 4 = ?$
A04	Horizontal; answer becomes 2 digits	$7 + 8 = ?$
A05	Addition involving zero; zero first or second	$0 + 6$; $6 + 0$

2.2 Two-digit + One-digit / One-digit + Two-digit

ID	Skill / Case	Example / diagnostic focus
A06	2-digit + 1-digit, vertical, no carry	$43 + 5$; align 5 under ones
A07	2-digit + 1-digit, horizontal, no carry	$43 + 5 = ?$
A08	1-digit + 2-digit, vertical, no carry	$5 + 43$; smaller number appears first
A09	1-digit + 2-digit, horizontal, no carry	$5 + 43 = ?$

ID	Skill / Case	Example / diagnostic focus
A10	2-digit + 1-digit, vertical, carry; answer 2 digits	$47 + 6$
A11	2-digit + 1-digit, horizontal, carry; answer 2 digits	$47 + 6 = ?$
A12	1-digit + 2-digit, vertical, carry; answer 2 digits	$6 + 47$
A13	1-digit + 2-digit, horizontal, carry; answer 2 digits	$6 + 47 = ?$
A14	2-digit + 1-digit, vertical, carry creates 3-digit answer	$96 + 8 = 104$
A15	2-digit + 1-digit, horizontal, carry creates 3-digit answer	$96 + 8 = ?$
A16	1-digit + 2-digit, horizontal/vertical, carry creates 3-digit answer	$8 + 96 = ?$

2.3 Two-digit + Two-digit

ID	Skill / Case	Example
A17	Vertical, no carry	$23 + 45$
A18	Horizontal, no carry	$23 + 45 = ?$
A19	Vertical, carry from ones only	$27 + 36 = 63$
A20	Horizontal, carry from ones only	$27 + 36 = ?$
A21	Vertical, carry from tens only; ones do not carry	$61 + 42 = 103$
A22	Horizontal, carry from tens only	$61 + 42 = ?$
A23	Vertical, carry from ones and tens	$68 + 47 = 115$
A24	Horizontal, carry from ones and tens	$68 + 47 = ?$

2.4 Three-digit + One-digit / One-digit + Three-digit

ID	Skill / Case	Example / diagnostic focus
A25	3-digit + 1-digit, vertical, no carry	$342 + 5$; 5 must be under ones
A26	3-digit + 1-digit, horizontal, no carry	$342 + 5 = ?$
A27	1-digit + 3-digit, vertical, no carry	$5 + 342$; order changes but alignment does not
A28	1-digit + 3-digit, horizontal, no carry	$5 + 342 = ?$
A29	3-digit + 1-digit, carry from ones only, vertical	$347 + 5 = 352$
A30	Same, horizontal	$347 + 5 = ?$
A31	1-digit first, carry from ones, vertical	$5 + 347$
A32	1-digit first, carry from ones, horizontal	$5 + 347 = ?$
A33	Carry cascades through one additional place	$399 + 4 = 403$
A34	Same cascade, horizontal	$399 + 4 = ?$
A35	Carry cascades through multiple 9s and creates new digit	$999 + 7 = 1006$
A36	Same, horizontal	$999 + 7 = ?$

2.5 Three-digit + Two-digit / Two-digit + Three-digit

ID	Skill / Case	Example / focus
A37	3-digit + 2-digit, vertical - No carry	$345 + 23$
A38	3-digit + 2-digit, horizontal - No carry	$345 + 23 = ?$
A39	2-digit + 3-digit, vertical - No carry	Shorter number first; preserve place alignment
A40	2-digit + 3-digit, horizontal - No carry	Shorter number first; convert correctly if needed
A41	3-digit + 2-digit, vertical - Carry from ones only	$347 + 25$
A42	3-digit + 2-digit, horizontal - Carry from ones only	$347 + 25 = ?$
A43	2-digit + 3-digit, vertical - Carry from ones only	Shorter number first; preserve place alignment
A44	2-digit + 3-digit, horizontal - Carry from ones only	Shorter number first; convert correctly if needed
A45	3-digit + 2-digit, vertical - Carry from tens only	$361 + 42 = 403$
A46	3-digit + 2-digit, horizontal - Carry from tens only	$361 + 42 = 403 = ?$
A47	2-digit + 3-digit, vertical - Carry from tens only	Shorter number first; preserve place alignment
A48	2-digit + 3-digit, horizontal - Carry from tens only	Shorter number first; convert correctly if needed
A49	3-digit + 2-digit, vertical - Carry from ones and tens	$378 + 47 = 425$
A50	3-digit + 2-digit, horizontal - Carry from ones and tens	$378 + 47 = 425 = ?$
A51	2-digit + 3-digit, vertical - Carry from ones and tens	Shorter number first; preserve place alignment
A52	2-digit + 3-digit, horizontal - Carry from ones and tens	Shorter number first; convert correctly if needed
A53	3-digit + 2-digit, vertical - Carry creates 4-digit answer	$985 + 27 = 1012$
A54	3-digit + 2-digit, horizontal - Carry creates 4-digit answer	$985 + 27 = 1012 = ?$
A55	2-digit + 3-digit, vertical - Carry creates 4-digit answer	Shorter number first; preserve place alignment
A56	2-digit + 3-digit, horizontal - Carry creates 4-digit answer	Shorter number first; convert correctly if needed

2.6 Three-digit + Three-digit - carry pattern matrix

ID	Carry pattern	Presentation
A57-A58	No carry	Vertical + Horizontal
A59-A60	Carry from ones only	Vertical + Horizontal
A61-A62	Carry from tens only	Vertical + Horizontal
A63-A64	Carry from hundreds only; answer gains a digit	Vertical + Horizontal
A65-A66	Carry from ones + tens	Vertical + Horizontal
A67-A68	Carry from ones + hundreds; non-adjacent carry	Vertical + Horizontal
A69-A70	Carry from tens + hundreds	Vertical + Horizontal
A71-A72	Carry from ones + tens + hundreds	Vertical + Horizontal

2.7 Addition involving zeros

Skill	Example	Distinct risk
Add zero to a number	$456 + 0$	Identity property / unnecessary manipulation
Zero first	$0 + 456$	Same concept with reversed order
Internal zero in one addend	$405 + 32$	Correct place alignment
Internal zeros in both addends	$405 + 302$	Column interpretation
Carry into a zero	$208 + 96$	Carry must be added to zero column
Carry through 9s ending at zero	$1099 + 8$	Cascade across places
Multiple zeros in large number	$2005 + 304$	Alignment and skipped places
Answer contains zeros	$495 + 505 = 1000$	Do not drop placeholder zeros

2.8 Addition of three or more numbers

Skill / Case	Example / focus
Three 1-digit addends	$8 + 7 + 9$; carry can be greater than 1
Three 2-digit addends, no carry	$21 + 34 + 12$
Three 2-digit addends, with carry	$78 + 67 + 89$
Three 3-digit addends	$245 + 378 + 126$
Unequal digit lengths, vertical	$345 + 27 + 6$; align all by ones place
Unequal digit lengths, horizontal	$345 + 27 + 6 = ?$
Four or more addends	$25 + 38 + 47 + 16$
Column sum produces carry 2 or more	$9 + 8 + 7 + 6 = 30$ in one column

2.9 N-digit addition - scalable cases

Case family	Assess separately
Digit-length combinations	$N+1$, $N+2$, ..., $N+N$; also shorter number first vs longer first
Presentation	Vertical vs horizontal
Place-value alignment	Equal lengths vs unequal lengths
Carry count	None / one / multiple
Carry pattern	Isolated / consecutive / non-adjacent / cascading through 9s
Answer size	Same number of digits vs one additional digit
Zeros	Internal zeros / multiple zeros / carry into zero / answer zeros

3. Subtraction - Direct Calculation Skills

3.1 Single-digit - Single-digit

Skill / Case	Example
Vertical; positive answer	$8 - 3$
Horizontal; positive answer	$8 - 3 = ?$
Answer is zero	$7 - 7 = 0$
Subtract zero	$7 - 0 = 7$

For a whole-number assessment, cases where the result would be negative should normally be excluded or treated as a separate integer-extension skill.

3.2 Two-digit - One-digit

Skill / Case	Example / focus
Vertical, no borrowing	$47 - 5$
Horizontal, no borrowing	$47 - 5 = ?$
Vertical, borrowing; answer 2 digits	$42 - 7 = 35$
Horizontal, borrowing; answer 2 digits	$42 - 7 = ?$
Vertical, borrowing; answer becomes 1 digit	$13 - 7 = 6$
Horizontal, borrowing; answer becomes 1 digit	$13 - 7 = ?$

3.3 Two-digit - Two-digit

Skill / Case	Example
Vertical, no borrowing; answer 2 digits	$76 - 23 = 53$
Horizontal, no borrowing; answer 2 digits	$76 - 23 = ?$
No borrowing; answer becomes 1 digit	$32 - 30 = 2$
No borrowing; answer zero	$44 - 44 = 0$
Vertical, borrow from tens; answer 2 digits	$72 - 38 = 34$
Horizontal, borrow from tens; answer 2 digits	$72 - 38 = ?$
Borrow from tens; answer becomes 1 digit	$43 - 38 = 5$

3.4 Three-digit - One-digit

Skill / Case	Example / focus
Vertical, no borrowing	$347 - 5$; 5 under ones
Horizontal, no borrowing	$347 - 5 = ?$
Borrow from tens	$342 - 7$
Borrow across a zero	$302 - 7$
Borrow through multiple zeros	$1000 - 7$
Answer loses a digit	$100 - 7 = 93$

3.5 Three-digit - Two-digit

Skill / Case	Example / focus
Vertical, no borrowing	$345 - 23$; align 23 under tens/ones
Horizontal, no borrowing	$345 - 23 = ?$
Borrow in ones column only	$354 - 27$
Borrow in tens column only	$352 - 170$
Borrow in both columns	$352 - 178$
Borrow across zero	$302 - 18$
Borrow through multiple zeros	$1000 - 28$
Answer remains 3 digits	$456 - 23 = 433$
Answer becomes 2 digits	$105 - 23 = 82$
Answer becomes 1 digit	$105 - 97 = 8$

3.6 Three-digit - Three-digit - borrow pattern matrix

Borrow pattern	Presentation / answer cases
No borrowing	Vertical + horizontal; answer may keep or lose digits
Borrow from tens only	Vertical + horizontal
Borrow from hundreds only	Vertical + horizontal
Borrow in ones and tens	Vertical + horizontal
Consecutive borrowing	Borrow required in adjacent columns
Borrow across one zero	e.g., $402 - 185$
Borrow across multiple zeros	e.g., $1000 - 476$
Internal zero in subtrahend	e.g., $764 - 203$

3.7 Subtraction involving zeros

Skill / Case	Example	Distinct risk
Number - 0	$456 - 0$	Identity property
Number - itself	$456 - 456$	Answer zero
Zero inside minuend	$504 - 203$	Borrow and place-value interpretation
Zero inside subtrahend	$764 - 203$	Do not treat zero as missing
Borrow through one zero	$402 - 185$	Regroup across an empty place
Borrow through multiple zeros	$1000 - 476$	Long borrow chain
Answer contains zero	$705 - 201 = 504$	Retain internal placeholder zero
Answer ends in zero	$650 - 120 = 530$	Retain trailing zero
Answer loses multiple digits	$1000 - 999 = 1$	Magnitude change

3.8 N-digit subtraction - scalable cases

Case family	Assess separately
Digit-length combinations	N-digit minus 1/2/.../N-digit; align by ones place
Presentation	Vertical vs horizontal
Borrow count	None / one / multiple
Borrow pattern	Isolated / consecutive / across one zero / across multiple zeros
Answer size	Same digits / one fewer digit / several fewer digits / zero
Zeros	Internal zeros in minuend or subtrahend; answer zeros

4. Place-Value Alignment & Horizontal-to-Vertical Conversion

These are independent diagnostic skills. A learner can know the arithmetic but still fail because the shorter number is placed under the wrong column.

Skill	Addition example	Subtraction example
Align 1-digit under ones of 2-digit number	$43 + 5$	$43 - 5$
Align 1-digit under ones of 3-digit number	$342 + 5$	$342 - 5$
Align 2-digit under tens/ones of 3-digit number	$345 + 23$	$345 - 23$
Align 1-digit under ones of 4+ digit number	$4,382 + 7$	$4,382 - 7$
Shorter number written first	$7 + 342$	Not applicable in ordinary whole-number subtraction unless result stays non-negative
Convert horizontal to vertical correctly	$345 + 27 = ?$	$345 - 27 = ?$
Preserve placeholder zeros while aligning	$4,005 + 73$	$4,005 - 73$

5. Regrouping Pattern Skills

5.1 Carry patterns in addition

Pattern	What to assess	Example type
No carry	Independent column addition	$243 + 514$
Single carry	One column generates a carry	$247 + 315$
Multiple independent carries	More than one column carries	$286 + 157$
Consecutive carries	Carry in adjacent columns	$678 + 457$
Non-adjacent carries	Carry in separated columns	Select digits so ones and hundreds carry but tens does not
Cascading carry	A carry triggers another carry because of 9s	$399 + 4$; $1,999 + 6$
Carry creates new leading digit	Highest place carries	$678 + 457 = 1,135$
Carry greater than 1	Three or more addends in same column	$8 + 7 + 9 = 24$

5.2 Borrow patterns in subtraction

Pattern	What to assess	Example type
No borrowing	Independent column subtraction	764 - 231
Single borrow	Borrow in one column	742 - 318
Multiple borrows	Borrow in more than one column	653 - 278
Consecutive borrowing	Adjacent columns both require borrowing	352 - 178
Borrow across one zero	Immediate left place is zero	402 - 185
Borrow across multiple zeros	Several zeros must be traversed	1000 - 476
Borrow changes another later decision	After borrowing, reduced digit may itself need to borrow	Typical multi-column regrouping
Highest-place reduction	Borrowing reduces leading digit and answer may lose a digit	105 - 97 = 8

6. Missing Number & Missing Digit Skills

6.1 Addition - whole addend missing

Skill	Example
Missing second addend, 1 digit	$5 + \square = 12$
Missing first addend, 1 digit	$\square + 5 = 12$
Missing 2-digit addend, no carry structure	$23 + \square = 58$
Missing 2-digit addend, carry structure	$47 + \square = 83$
Missing 3-digit addend	$245 + \square = 613$
Missing addend in 3+ addend expression	$35 + \square + 18 = 80$

6.2 Addition - digit(s) missing inside numbers

Skill	Example
One missing digit, no carry	$2\square + 34 = 59$
One missing digit, carry required	$4\square + 38 = 85$
Missing tens digit	$\square 7 + 25 = 62$
Missing ones digit	$4\square + 27 = 72$
Missing hundreds digit	$\square 24 + 352 = 876$
Missing digit in answer	$37 + 46 = 8\square$
Two missing digits across addends	$4\square + \square 3 = 78$
Missing digits in addends and answer	$\square 4\square + 2\square 5 = 6\square 9$
Same symbol means same digit	$A5 + 2A = 88$
Find all possible digits under inequality constraint	$3\square + 27 < 70$

6.3 Subtraction - whole number missing

Unknown position	Example	Reasoning demand
Difference missing	$15 - 7 = \square$	Direct calculation
Subtrahend missing	$15 - \square = 7$	Reverse reasoning
Minuend missing	$\square - 7 = 8$	Reverse reasoning / inverse operation

Unknown position	Example	Reasoning demand
2-digit subtrahend missing	$58 - \square = 24$	Whole-number unknown
3-digit minuend missing	$\square - 275 = 418$	Inverse reasoning at larger magnitude

6.4 Subtraction - digit(s) missing

Skill	Example
One missing digit without borrowing	$5\square - 23 = 34$
One missing digit requiring borrowing	$6\square - 28 = 35$
Missing digit in minuend	$\square 7 - 29 = 38$
Missing digit in subtrahend	$64 - 2\square = 37$
Missing digit in answer	$74 - 28 = 4\square$
Missing digit with zero/borrow chain	$4\square 2 - 185 = 217$
Multiple missing digits	$\square 5 - 2\square = 17$
Same symbol means same digit	$A5 - 2A = 27$
All possible solutions under constraint	$7\square - 25 > 50$

7. Missing Operation, Equality & Inverse Relationship

Skill family	Cases / examples
Missing operation	$7 \square 3 = 10$; $15 \square 8 = 7$; choose + or -
Multiple missing operations	$12 \square 5 \square 3 = 10$
Fact family from addition	$7 + 5 = 12 \rightarrow 5 + 7 = 12$; $12 - 7 = 5$; $12 - 5 = 7$
Related addition fact from subtraction	Given $14 - 6 = 8$, write $8 + 6 = 14$
Check addition using subtraction	If $347 + 258 = 605$, verify $605 - 258 = 347$
Check subtraction using addition	If $605 - 258 = 347$, verify $347 + 258 = 605$
Balanced equation - one blank	$12 + 8 = 15 + \square$
Balanced equation - different operations	$43 - 7 = \square + 11$
Blank on both sides	$25 + \square = 40 - \square$
True / false equality	$35 + 17 = 40 + 12?$
Compare without calculating fully	$198 + 47$ vs $200 + 45$

8. Efficient / Mental Addition & Subtraction

This section measures strategy choice, not only whether the final answer is correct.

Skill	Example / strategy
Add/subtract 10, 100, 1000	$347 + 100$; $812 - 100$
Make 10	$8 + 7 \rightarrow 8 + 2 + 5$
Make 100 / 1000	$68 + 32$; $650 + 350$
Compensation for addition	$56 + 29 \rightarrow 56 + 30 - 1$
Compensation for subtraction	$83 - 39 \rightarrow 83 - 40 + 1$
Near hundred / thousand	$298 + 154$; $1002 - 398$

Reorder using commutativity	$27 + 36 + 13 \rightarrow (27+13)+36$
Group convenient pairs	$25 + 37 + 75 \rightarrow (25+75)+37$
Break apart by place value	$347 + 25 \rightarrow 347 + 20 + 5$
Count on for small difference	503 - 498 by counting from 498 to 503
Use inverse operation	Find $\square$ in $47 + \square = 83$ via $83 - 47$
Choose efficient method	$199 + 47$ vs standard column algorithm

9. Estimation, Reasonableness & Checking

Skill	Example / purpose
Estimate a sum by rounding	$398 + 207 \approx 400 + 200$
Estimate a difference by rounding	$802 - 396 \approx 800 - 400$
Choose closest estimate	Which is closest to $487 + 216$?
Detect impossible answer by magnitude	$487 + 216$ cannot be 6,903
Check final digit / parity clue	Even + odd must be odd; useful as a reasonableness check
Use inverse to verify	Addition checked by subtraction; subtraction checked by addition
Compare exact answer with estimate	Does 704 make sense for $398 + 307$?

10. Addition & Subtraction Word-Problem Skills

10.1 One-step structures

Structure	Addition/Subtraction role	Example prompt pattern
Result unknown - join	Addition	Had 24, got 13 more. How many now?
Change unknown - join	Addition/subtraction inverse	Had 24, now 37. How many were added?
Start unknown - join	Subtraction/inverse	Had some, got 13, now 37. How many initially?
Result unknown - separate	Subtraction	Had 37, gave away 13. How many left?
Change unknown - separate	Subtraction/inverse	Had 37, now 24. How many were given away?
Start unknown - separate	Addition/inverse	Had some, gave away 13, now 24. How many initially?
Part-part-whole - whole unknown	Addition	24 boys and 18 girls. Total?
Part-part-whole - part unknown	Subtraction	42 total, 24 boys. Girls?
Compare - difference unknown	Subtraction	Riya has 42; Aman has 35. How many more?
Compare - larger unknown	Addition	Aman has 35; Riya has 7 more. How many Riya?
Compare - smaller unknown	Subtraction	Riya has 42; Aman has 7 fewer. How many Aman?

10.2 Language and representation variations

Skill	Examples / notes
Recognise addition language	total, altogether, in all, combined, increased by, more added
Recognise subtraction language	left, remaining, difference, fewer, how many more, decreased by
Solve without relying on keywords	Operation must be inferred from situation, not one word
Translate story into number sentence	Write $48 - 19 = ?$ before solving

Skill	Examples / notes
Use irrelevant information correctly	Ignore data not needed to answer
Interpret tables/charts/simple data	Add or subtract values read from a representation
Missing start / change / result	Unknown can appear in any position
Same context, different question	Ask total vs difference vs missing part using same numbers

10.3 Multi-step word problems

Skill	Structure
Addition then addition	Two increases / combine three quantities
Subtraction then subtraction	Two decreases
Addition then subtraction	Quantity increases, then decreases
Subtraction then addition	Quantity decreases, then increases
Choose order of operations	Correct sequence matters
Intermediate result required	Second step depends on first
Unknown in first step	Work backwards
Unknown in second step	Forward then reverse reasoning
Extra information included	Select relevant facts
Olympiad-style constraint story	Several conditions; may require trying possibilities or inverse reasoning

11. Error-Diagnosis Skills

Error type to test	Typical learner error
Place-value misalignment	Writes 5 under tens in $342 + 5$
Horizontal-to-vertical conversion error	Shifts a shorter number left
Forgotten carry	Adds column but does not transfer carry
Carry placed in wrong column	Writes carry above wrong place
Carry not included	Records carry but forgets to add it
Assumes carry is always 1	Fails when 3+ addends create carry 2 or more
Drops zero placeholder	Writes $495 + 505$ as 100 instead of 1000
Subtracts smaller digit from larger regardless of position	In $52 - 38$, computes 8-2 instead of regrouping
Forgotten borrow	Attempts 2-8 directly without regrouping
Does not reduce donor digit	Borrows 10 but leaves original tens unchanged
Borrow from wrong place	Incorrect regrouping source
Borrow across zero incorrectly	Fails on $402 - 185$ or $1000 - 476$
Drops leading/trailing/internal zero in answer	Place-value representation error
Reverses subtraction	Treats subtraction as commutative
Incorrect missing-digit logic	Finds a digit that works locally but not across carries/borrows
Equality misconception	Treats = as “the answer comes next” rather than balance

12. Master Tagging Matrix for an Assessment Bank

Tag	Allowed values / examples
operation	ADD / SUB
operand_1_digits	1, 2, 3, 4, ..., N
operand_2_digits	1, 2, 3, 4, ..., N
operand_order	LONGER_FIRST / SHORTER_FIRST / EQUAL_LENGTH
presentation	VERTICAL / HORIZONTAL / STORY / EQUATION_BALANCE
alignment_required	YES / NO
regrouping	NONE / SINGLE / MULTIPLE
regroup_columns	ONES / TENS / HUNDREDS / ...
regroup_pattern	ISOLATED / CONSECUTIVE / NON_ADJACENT / CASCADING / ACROSS_ZERO
zero_pattern	NONE / INTERNAL / MULTIPLE / ANSWER_ZERO
answer_digit_change	SAME / +1 / -1 / -MULTIPLE / ZERO
num_operands	2 / 3 / 4+
unknown_type	NONE / WHOLE_NUMBER / DIGIT / MULTIPLE_DIGITS / OPERATION
unknown_position	FIRST_OPERAND / SECOND_OPERAND / RESULT / MULTIPLE
reasoning_type	DIRECT / INVERSE / BALANCE / CONSTRAINT / ERROR_DIAGNOSIS
strategy	STANDARD / MENTAL / COMPENSATION / ESTIMATION
context	BARE_NUMBER / WORD_PROBLEM / TABLE_OR_CHART

Recommended database rule: do not encode every possible combination as a separate hard-coded topic name. Store these dimensions as tags, then generate or select question sets by combinations of tags. This prevents cases such as “3-digit + 1-digit, horizontal, carry” from being accidentally omitted.

13. Suggested Assessment Progression

Stage	What it should establish
1. Direct, no regrouping	Basic operation and place-value understanding
2. Horizontal vs vertical	Representation flexibility and correct alignment
3. Unequal digit lengths	Alignment of ones/tens/hundreds
4. Single regrouping	Carry/borrow mechanism
5. Multiple/cascading regrouping	Robust column procedure
6. Zeros	Regrouping through empty places / placeholder understanding
7. Answer-size changes	Magnitude awareness
8. Three or more addends	Higher column totals and carry >1
9. Missing number / missing digit	Reverse reasoning and Olympiad-style constraints
10. Equality / inverse / checking	Structural understanding of operations
11. Efficient mental strategies	Fluency beyond standard algorithm
12. Word problems	Application and operation selection
13. Error diagnosis	Ability to detect and explain misconceptions

End of taxonomy